sMDD (k = k*) vs MDD — DD Python, paired rows (same instance, DD type, width and reduction)
MDD: results_python_01_11_2026_17-48-09.csv | sMDD: results_python_semi_kopt_06_11_2026_07-34-11.csv
Columns are means over the paired rows; 'ratio' is the geometric mean of sMDD/MDD per row. Time = construction + reduction; Solve = longest/shortest path over the built diagram.
Bound means skip rows where either value is ±inf (empty restricted / unreachable terminal), counted in 'non-fin.'. 'sMDD +/=/-': rows where the sMDD bound is better / equal / worse.

Problem	DD	Red.	N	Time MDD	Time sMDD	ratio	Solve MDD	Solve sMDD	ratio	Nodes MDD	Nodes sMDD	ratio	Arcs MDD	Arcs sMDD	ratio	Bound MDD	Bound sMDD	non-fin.	sMDD +/=/-
Knapsack	Exact	no	24	31.21	48.40	1.56×	11.14	18.25	1.42×	3.26e+06	3.21e+06	0.58×	6.28e+06	2.25e+07	2.73×	13,215	13,215	0	0/24/0
	Exact	yes	21	9.49	17.63	1.85×	2.55	3.97	1.69×	872,545	872,545	1.00×	1.64e+06	6.17e+06	3.20×	7,599	7,599	0	0/21/0
	Restricted	no	147	9.22	12.35	1.38×	3.12	3.77	1.34×	1.21e+06	1.18e+06	0.59×	2.09e+06	4.97e+06	2.16×	11,275	11,275	0	0/147/0
	Restricted	yes	147	11.17	16.93	1.60×	3.01	3.46	1.46×	1.17e+06	1.17e+06	0.97×	2.05e+06	4.95e+06	2.50×	11,275	11,275	0	0/147/0
	RelaxPriority	no	147	10.53	11.13	1.25×	2.83	3.15	1.21×	1.21e+06	1.19e+06	0.60×	2.1e+06	4.11e+06	1.88×	300,155	299,432	0	46/101/0
	RelaxPriority	yes	147	12.66	14.71	1.39×	2.82	3.30	1.30×	1.17e+06	1.16e+06	0.96×	2.05e+06	4.05e+06	2.10×	300,155	299,432	0	46/101/0
Independent Set	Exact	no	14	6.86	94.08	1.80×	1.58	1.19	0.60×	872,766	777,701	0.64×	1.07e+06	1.31e+06	1.01×	27.93	27.93	0	0/14/0
	Exact	yes	14	7.91	95.51	1.66×	1.57	1.21	0.63×	872,766	777,701	0.64×	1.07e+06	1.31e+06	1.01×	27.93	27.93	0	0/14/0
	Restricted	no	117	1.32	17.17	2.54×	0.2808	0.3533	0.84×	176,224	151,793	0.79×	202,300	735,314	1.25×	23.88	22.90	0	0/93/24
	Restricted	yes	117	1.53	17.46	2.37×	0.2867	0.3650	0.85×	174,694	149,528	0.79×	200,283	728,053	1.24×	23.88	22.90	0	0/93/24
	RelaxPriority	no	118	1.91	18.30	2.09×	0.3341	0.4727	0.85×	234,074	224,572	0.84×	267,211	1.02e+06	1.22×	70.91	77.49	0	17/63/38
	RelaxPriority	yes	118	2.35	18.78	1.92×	0.3602	0.1732	0.52×	233,953	107,250	0.58×	267,859	227,277	0.75×	71.14	77.88	0	16/63/39
Set Cover	Exact	no	190	0.8765	71.09	0.77×	0.1521	11.57	1.46×	86,373	62,827	0.95×	166,820	2.47e+07	1.84×	3.63	3.63	0	0/190/0
	Exact	yes	188	0.8939	68.03	0.79×	0.0973	3.91	1.33×	54,687	50,218	0.99×	107,873	9.05e+06	1.70×	3.62	3.62	0	0/188/0
	Restricted	no	1234	0.4625	4.88	0.69×	0.0726	0.5361	1.22×	48,117	49,066	1.01×	93,561	1.59e+06	1.52×	3.81	3.81	0	0/1234/0
	Restricted	yes	1230	0.5145	5.54	0.73×	0.0629	0.2797	1.11×	40,741	43,458	1.02×	80,008	798,878	1.39×	3.81	3.81	0	0/1230/0
	RelaxPriority	no	1234	0.4952	5.27	0.70×	0.0727	0.5370	1.21×	48,272	51,006	1.02×	94,829	1.63e+06	1.53×	3.16	3.16	0	11/1205/18
	RelaxPriority	yes	1230	0.5497	5.95	0.74×	0.0646	0.2785	1.11×	41,183	43,744	1.02×	81,412	791,276	1.39×	3.16	3.15	0	8/1204/18
SOC Knapsack	Exact	no	84	4.16	44.88	2.86×	0.9679	7.99	2.13×	415,294	9,576	0.07×	706,467	1.73e+07	5.89×	791	791	0	0/84/0
	Exact	yes	84	5.18	69.78	3.57×	0.0215	5.76	25.51×	12,070	8,067	0.92×	23,577	1.28e+07	44.42×	791	791	0	0/84/0
	Restricted	no	530	0.2397	7.43	6.18×	0.0436	0.7748	4.43×	32,318	6,251	0.21×	39,681	2.4e+06	11.02×	688	771	0	351/179/0
	Restricted	yes	527	0.2666	8.58	7.41×	0.0011	0.3703	26.89×	1,579	4,203	1.83×	2,705	1.19e+06	45.35×	687	770	0	348/179/0
	RelaxPriority	no	531	1.17	7.61	3.24×	0.0550	0.6874	2.69×	35,509	8,069	0.22×	51,861	2.23e+06	5.68×	892	869	0	226/305/0
	RelaxPriority	yes	528	1.15	8.33	3.96×	0.0015	0.3413	10.28×	1,860	2,841	1.04×	3,379	1.08e+06	15.45×	891	868	0	224/304/0
Sequencing	Exact	no	51	4.51	0.4099	0.20×	1.89	0.5804	0.40×	125,204	16,373	0.22×	898,317	125,213	0.24×	1,672	1,672	0	0/51/0
	Exact	yes	51	5.76	0.4801	0.18×	0.6226	0.5526	1.00×	16,373	16,373	1.00×	125,213	125,213	1.00×	1,672	1,672	0	0/51/0
	Restricted	no	396	1.57	1.29	0.51×	0.4398	0.8235	1.01×	36,373	24,154	0.49×	201,912	162,268	0.58×	2,201	2,175	0	121/275/0
	Restricted	yes	396	1.75	1.43	0.47×	0.1515	0.8109	2.48×	6,709	24,154	1.96×	36,383	162,268	2.23×	2,201	2,175	0	121/275/0
	RelaxPriority	no	396	4.50	4.63	0.53×	1.25	2.55	0.94×	46,290	32,201	0.49×	636,438	435,183	0.47×	1,336	1,142	0	66/156/174
	RelaxPriority	yes	396	5.09	4.86	0.48×	0.3674	2.55	2.51×	7,537	32,201	2.14×	100,671	435,183	1.96×	1,323	1,142	0	66/156/174